BrightLearn
AI & DATA SKILLS

STUDENT EXERCISE · SQL DATE FUNCTIONS

# SQL Date Functions & Date Math

A hands-on, pen-and-paper practice exercise. Read each table, write the SQL query that returns the requested data, and draw the expected output table.

**COURSE**
BrightLearn Basic SQL

**FORMAT**
Handwritten · Pen & Paper

**QUESTIONS**
20 questions + 1 bonus

**FOCUS**
Extract · Calculate · Format



Extract · Calculate · Format
Learn SQL by writing it.

# 00 Exercise Instructions

This exercise focuses on the date functions used most often in analytics — extracting day, month, year and quarter values, calculating the difference between two dates, and formatting dates for reporting. Work through each question carefully — accuracy beats speed.

### ★ LEARNING OBJECTIVES

By the end of this exercise you will be able to:

- Extract date parts with DAYNAME, MONTHNAME, MONTH, YEAR, DAY and QUARTER.
- Use CURRENT_DATE, TO_DATE and TO_CHAR to convert and format date values.
- Calculate differences and add intervals with DATEDIFF and DATEADD.
- Find period boundaries with DATE_TRUNC and LAST_DAY.
- Filter rows by year, month or day difference and classify dates with CASE.

### FOR EACH QUESTION, DO THIS:

**1**

**WRITE THE QUERY**
Hand-write the full SQL statement that produces the requested result. Use SQL keywords in CAPS.

**2**

**DRAW THE OUTPUT**
Draw the expected output table by hand — column headers on top, rows underneath.

**3**

**CHECK COLUMNS**
Make sure your output uses the exact column names listed under Expected Columns.

### SUBMISSION RULES

- **Format:** All answers must be handwritten on lined paper, using pen (no pencil, no typing).
- **Layout:** Number every answer to match the question (1 through 20, plus the bonus).
- **Queries:** Write each SQL statement neatly. SQL keywords in UPPERCASE, table and column names in lowercase.
- **Output tables:** Draw clear column headers, then list each row beneath. Use a ruler if your handwriting needs help staying straight.
- **Originality:** All work must be your own — this exercise is graded on your understanding, not a perfect answer.

# 01 Questions 1–5 — Extract Date Parts

Extract individual pieces from a date column — the day name, month name, month number, year, and day of the month.

## 01 Day Name from Order Date

TABLE orders

| order_id | customer_id | order_date |
|---|---|---|
| 1001 | 101 | 2026-05-01 |
| 1002 | 102 | 2026-05-02 |
| 1003 | 103 | 2026-05-03 |
| 1004 | 104 | 2026-05-04 |
| 1005 | 105 | 2026-05-05 |

TASK

Return each order with the day name of the order date.

USE

DAYNAME AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date day_name

WRITE YOUR SQL HERE (handwritten)

## 02 Month Name from Signup Date

TABLE customer_signups

| customer_id | customer_name | signup_date |
|---|---|---|
| 201 | John | 2026-01-15 |
| 202 | Mary | 2026-02-20 |
| 203 | Peter | 2026-03-05 |
| 204 | Sarah | 2026-04-18 |
| 205 | Thabo | 2026-05-30 |

TASK

Return each customer with the name of the month in which they signed up.

USE

MONTHNAME AS

EXPECTED OUTPUT COLUMNS

customer_id customer_name signup_date signup_month_name

WRITE YOUR SQL HERE ( handwritten )

## 03 Month Number from Sale Date

TABLE sales

| sale_id | product_name | sale_date | amount |
|---|---|---|---|
| S001 | Laptop | 2026-01-10 | 12000 |
| S002 | Mouse | 2026-02-15 | 250 |
| S003 | Keyboard | 2026-03-20 | 700 |
| S004 | Monitor | 2026-04-25 | 3500 |
| S005 | Desk | 2026-05-30 | 2000 |

TASK

Return each sale with the month number extracted from sale_date.

USE

`MONTH` `AS`

EXPECTED OUTPUT COLUMNS

`sale_id` `product_name` `sale_date` `sale_month`

WRITE YOUR SQL HERE (handwritten)

## 04 Year from Transaction Date

TABLE transactions

| transaction_id | customer_id | transaction_date | amount |
|---|---|---|---|
| T001 | 101 | 2024-12-15 | 500 |
| T002 | 102 | 2025-01-20 | 1200 |
| T003 | 103 | 2025-06-10 | 800 |
| T004 | 104 | 2026-02-05 | 1500 |
| T005 | 105 | 2026-05-25 | 2000 |

TASK

Return each transaction with the year extracted from transaction_date.

USE

YEAR AS

EXPECTED OUTPUT COLUMNS

transaction_id customer_id transaction_date transaction_year

WRITE YOUR SQL HERE (handwritten)

## 05 Day of Month from Delivery Date

TABLE deliveries

| delivery_id | customer_id | delivery_date |
|---|---|---|
| D001 | 101 | 2026-05-01 |
| D002 | 102 | 2026-05-08 |
| D003 | 103 | 2026-05-15 |
| D004 | 104 | 2026-05-22 |
| D005 | 105 | 2026-05-29 |

TASK

Return each delivery with the day of the month extracted from delivery_date.

USE

DAY AS

EXPECTED OUTPUT COLUMNS

delivery_id customer_id delivery_date day_of_month

WRITE YOUR SQL HERE (handwritten)

# 02 Questions 6–10 — Current Date, Convert & Format

Use the current date, convert text strings to date values, and format dates for display.

## 06 Add Today's Date

TABLE employees

| employee_id | employee_name | department |
|---|---|---|
| 301 | Nandi | Sales |
| 302 | Brian | IT |
| 303 | Lerato | Finance |
| 304 | Sipho | HR |
| 305 | Aisha | Marketing |

TASK
Return all employees and add a column showing today's date.

USE
`CURRENT_DATE()` `AS`

EXPECTED OUTPUT COLUMNS
`employee_id` `employee_name` `department` `today_date`

WRITE YOUR SQL HERE (handwritten)

## 07 Convert Text to Date

TABLE online_orders

| order_id | customer_id | order_date_text |
|---|---|---|
| 4001 | 101 | 2026-01-15 |
| 4002 | 102 | 2026-02-20 |
| 4003 | 103 | 2026-03-25 |
| 4004 | 104 | 2026-04-10 |
| 4005 | 105 | 2026-05-05 |

TASK

Convert the order_date_text string column into a proper date column.

USE

TO_DATE AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date_text order_date

WRITE YOUR SQL HERE (handwritten)

## 08 Format Date as Text

TABLE payment_dates

| payment_id | customer_id | payment_date |
|---|---|---|
| P001 | 101 | 2026-01-05 |
| P002 | 102 | 2026-02-10 |
| P003 | 103 | 2026-03-15 |
| P004 | 104 | 2026-04-20 |
| P005 | 105 | 2026-05-25 |

TASK

Format payment_date as a text string in the format YYYY-MM-DD.

USE

TO_CHAR AS

EXPECTED OUTPUT COLUMNS

payment_id customer_id payment_date formatted_payment_date

WRITE YOUR SQL HERE (handwritten)

## 09 Days Since Last Purchase

TABLE customer_purchases

| customer_id | customer_name | last_purchase_date |
|---|---|---|
| 501 | John | 2026-05-01 |
| 502 | Mary | 2026-05-10 |
| 503 | Peter | 2026-05-15 |
| 504 | Sarah | 2026-05-20 |
| 505 | Thabo | 2026-05-25 |

TASK

Calculate how many days have passed since each customer's last purchase. Use today's date.

USE

CURRENT_DATE() DATEDIFF AS

EXPECTED OUTPUT COLUMNS

customer_id customer_name last_purchase_date days_since_last_purchase

WRITE YOUR SQL HERE (handwritten)

## 10 Expected Delivery Date

TABLE shipping_orders

| order_id | customer_id | order_date |
|---|---|---|
| 6001 | 101 | 2026-05-01 |
| 6002 | 102 | 2026-05-03 |
| 6003 | 103 | 2026-05-05 |
| 6004 | 104 | 2026-05-07 |
| 6005 | 105 | 2026-05-09 |

TASK

Calculate the expected delivery date by adding 7 days to the order_date.

USE

DATEADD AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date expected_delivery_date

WRITE YOUR SQL HERE (handwritten)

# 03 Questions 11–15 — Multi-Extract & Date Filtering

Extract multiple date parts at once, filter rows by year or month, and use LAST_DAY and DATE_TRUNC.

## 11 Year, Month, Day Separately

TABLE bookings

| booking_id | customer_id | booking_date |
|---|---|---|
| B001 | 101 | 2026-01-12 |
| B002 | 102 | 2026-02-18 |
| B003 | 103 | 2026-03-22 |
| B004 | 104 | 2026-04-09 |
| B005 | 105 | 2026-05-27 |

TASK

Extract the year, month number, and day from booking_date into three separate columns.

USE

YEAR MONTH DAY AS

EXPECTED OUTPUT COLUMNS

booking_id customer_id booking_date booking_year booking_month booking_day

WRITE YOUR SQL HERE (handwritten)

## 12 Filter Orders from 2026

TABLE yearly_orders

| order_id | customer_id | order_date | amount |
|---|---|---|---|
| 7001 | 101 | 2024-12-15 | 500 |
| 7002 | 102 | 2025-01-20 | 1200 |
| 7003 | 103 | 2025-06-10 | 800 |
| 7004 | 104 | 2026-02-05 | 1500 |
| 7005 | 105 | 2026-05-25 | 2000 |

TASK

Return only orders from the year 2026.

USE

YEAR WHERE AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date order_year amount

WRITE YOUR SQL HERE (handwritten)

## 13 Filter Orders from March

TABLE monthly_orders

| order_id | customer_id | order_date | amount |
| --- | --- | --- | --- |
| 8001 | 101 | 2026-01-15 | 500 |
| 8002 | 102 | 2026-02-20 | 1200 |
| 8003 | 103 | 2026-03-10 | 800 |
| 8004 | 104 | 2026-03-25 | 1500 |
| 8005 | 105 | 2026-05-30 | 2000 |

TASK

Return only orders placed in March (month number 3).

USE

MONTH WHERE AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date order_month amount

WRITE YOUR SQL HERE (handwritten)

## 14 Last Day of the Month

TABLE subscriptions

| subscription_id | customer_id | start_date |
|---|---|---|
| SUB001 | 101 | 2026-01-10 |
| SUB002 | 102 | 2026-02-15 |
| SUB003 | 103 | 2026-03-20 |
| SUB004 | 104 | 2026-04-25 |
| SUB005 | 105 | 2026-05-30 |

TASK

Return the last day of the month for each subscription start date.

USE

LAST_DAY AS

EXPECTED OUTPUT COLUMNS

subscription_id customer_id start_date month_end_date

WRITE YOUR SQL HERE (handwritten)

## 15 First Day of the Month

TABLE campaign_sends

| send_id | customer_id | send_date |
|---|---|---|
| C001 | 101 | 2026-01-12 |
| C002 | 102 | 2026-02-18 |
| C003 | 103 | 2026-03-22 |
| C004 | 104 | 2026-04-09 |
| C005 | 105 | 2026-05-27 |

TASK

Return the first day of the month for each campaign send date.

USE

DATE_TRUNC AS

EXPECTED OUTPUT COLUMNS

send_id customer_id send_date month_start_date

WRITE YOUR SQL HERE ( handwritten )

# 04 Questions 16–20 — Advanced Date Work

The final five questions involve formatting, age calculations, CASE with dates, quarter extraction, and filtering by date difference.

## 16 Format as Month and Year

TABLE invoice_dates

| invoice_id | customer_id | invoice_date |
|---|---|---|
| INV001 | 101 | 2026-01-05 |
| INV002 | 102 | 2026-02-10 |
| INV003 | 103 | 2026-03-15 |
| INV004 | 104 | 2026-04-20 |
| INV005 | 105 | 2026-05-25 |

TASK

Format invoice_date as a text string showing month name and year. Example: January 2026.

USE

TO_CHAR AS

EXPECTED OUTPUT COLUMNS

invoice_id customer_id invoice_date invoice_month_year

WRITE YOUR SQL HERE (handwritten)

## 17 Customer Age from Date of Birth

TABLE customer_birthdays

| customer_id | customer_name | date_of_birth |
|---|---|---|
| 901 | John | 1998-05-10 |
| 902 | Mary | 1990-08-20 |
| 903 | Peter | 2002-03-15 |
| 904 | Sarah | 1985-12-01 |
| 905 | Thabo | 2000-07-30 |

TASK

Calculate each customer’s age in years using their date of birth and today’s date.

USE

CURRENT_DATE() DATEDIFF or YEAR AS

EXPECTED OUTPUT COLUMNS

customer_id customer_name date_of_birth customer_age

WRITE YOUR SQL HERE (handwritten)

## 18 Classify Weekend vs Weekday

TABLE weekend_orders

| order_id | customer_id | order_date |
|---|---|---|
| 9001 | 101 | 2026-05-01 |
| 9002 | 102 | 2026-05-02 |
| 9003 | 103 | 2026-05-03 |
| 9004 | 104 | 2026-05-04 |
| 9005 | 105 | 2026-05-05 |

TASK

Classify each order as Weekend or Weekday based on the day name of the order date. Saturday and Sunday = Weekend. All others = Weekday.

USE

DAYNAME CASE AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date day_name day_type

WRITE YOUR SQL HERE (handwritten)

## 19 Transactions by Quarter

TABLE quarterly_transactions

| transaction_id | customer_id | transaction_date | amount |
|---|---|---|---|
| Q001 | 101 | 2026-01-15 | 500 |
| Q002 | 102 | 2026-03-20 | 1200 |
| Q003 | 103 | 2026-04-10 | 800 |
| Q004 | 104 | 2026-07-05 | 1500 |
| Q005 | 105 | 2026-10-25 | 2000 |

TASK

Extract the quarter number from each transaction date.

USE

QUARTER AS

EXPECTED OUTPUT COLUMNS

transaction_id customer_id transaction_date transaction_quarter amount

WRITE YOUR SQL HERE (handwritten)

## 20 Orders Older Than 30 Days

TABLE recent_orders

| order_id | customer_id | order_date | amount |
|---|---|---|---|
| R001 | 101 | 2026-04-01 | 500 |
| R002 | 102 | 2026-04-15 | 1200 |
| R003 | 103 | 2026-05-01 | 800 |
| R004 | 104 | 2026-05-10 | 1500 |
| R005 | 105 | 2026-05-25 | 2000 |

TASK

Return only orders that are more than 30 days old from today. Calculate days_since_order using today's date.

USE

CURRENT_DATE() DATEDIFF WHERE AS

EXPECTED OUTPUT COLUMNS

order_id customer_id order_date days_since_order amount

WRITE YOUR SQL HERE (handwritten)

# 05 Bonus Challenge

The bonus combines DATEDIFF, CASE, and CURRENT_DATE into a single customer segmentation query. Apply every concept from this guide in one question.

## B Customer Recency Segment

TABLE customer_recency

| customer_id | customer_name | last_purchase_date | total_spend |
|---|---|---|---|
| 1001 | John | 2026-05-25 | 5000 |
| 1002 | Mary | 2026-05-10 | 2500 |
| 1003 | Peter | 2026-04-01 | 700 |
| 1004 | Sarah | 2026-02-15 | 15000 |
| 1005 | Thabo | 2025-12-20 | 300 |

TASK

Calculate days_since_last_purchase for each customer using CURRENT_DATE. Then use CASE to classify: ≤ 30 days = Active Customer, 31–90 days = At Risk Customer, > 90 days = Inactive Customer.

USE

CURRENT_DATE() DATEDIFF CASE AS

EXPECTED OUTPUT COLUMNS

customer_id customer_name last_purchase_date days_since_last_purchase customer_status

WRITE YOUR SQL HERE (handwritten)

# Extract it.
# Calculate it.

Master SQL date functions by writing real queries on twenty business tables — from extracting day names to calculating customer recency segments.

★ BRIGHTLEARN · SQL DATE FUNCTIONS ★